Fiscal Year:	FY 2019	**Task Last Updated:**	FY 08/03/2019
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Project Title:	High-Precision Microwave Spectroscopy of Long-Lived Circular-State Rydberg Atoms in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48109-1040	**Congressional District:**	12
Comments:			
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
Contact Email:	ulf.e.israelsson@jpl.nasa.gov		
Flight Program:	ISS		
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Key Personnel Changes/Previous PI:	2019 report: No changes.		
COI Name (Institution):	Malinovsky, Vladimir Ph.D. (Stevens Institute of Technology)		
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Task Description:	Objective: The system of fundamental constants (the speed of light, Planck's constant, the fine structure constant, the Rydberg constant, etc.) is important because it directly reflects our current understanding of the foundations of nature. The objective of the project is to measure one of these constants, the Rydberg constant, at an uncertainty level in the range of a few parts in 10^12 (the current uncertainty stands at 5.9 parts in 10^12). It is planned to use circular Rydberg atoms, which are very large, highly excited atoms with doughnut-shaped electron distributions that do not penetrate into the atomic nucleus. As a result, these atoms are not susceptible to perturbations caused by electron wavefunction overlap with the nuclear charge density and quantum-electrodynamics (QED) corrections. The measurement therefore yields a result for the Rydberg constant that is not contingent to the accuracy/precision of input from other areas of physics. It is this property that makes a measurement of the Rydberg constant with circular Rydberg atoms particularly useful, as it will present a valuable complement/check for results based on established methods. Due to the absence of critical QED corrections and nuclear penetration, a measurement of the Rydberg constant with circular Rydberg atoms would help resolve a recently reported inconsistency, where it was concluded that "... either the Rydberg constant has to be shifted by -110kHz/c (4.9 standard deviations), or the calculations of the QED [quantum-electro-dynamic] effects in atomic hydrogen or muonic hydrogen atoms are insufficient." [R. Pohl et al., Nature 466, 213 (2010)]. Method: Rb atoms are cooled to below 1 microK and laser-excited into a low-angular-momentum Rydberg level with principal quantum number n on the order of 40. Using an adiabatic atomic-state transformation method, the Rydberg atoms are then transferred into the circular state |n,n1=0,n2=0,m=n-1> (m, n1, and n2 are the magnetic and parabolic quantum numbers). An optical lattice (wavelength 1064 nm or 532 nm on the ground, 850 nm at CAL) is adiabatically ramped on. The Rydberg atoms are trapped in the lattice via the ponderomotive interaction of the Rydberg electron with the lattice light. During the hold time of the atom cloud in the lattice, the electric-quadrupole transition into the state |n+2,n1=1,n2=1,m=n-1> is driven by amplitude-modulating the optical lattice at the transition frequency. The transition is probed using state-selective electric-field ionization. The measured transition frequency allows one to extract the Rydberg constant. Key advantages of the method are that the transition is free of first-order Stark and Zeeman shifts, that QED, nuclear-penetration, hyperfine, and other shifts are small, and that the optical lattice causes only moderate trap-induced transition shifts. Theoretical work includes the modeling of the circular-state preparation using multi-level adiabatic passage and of the circular-state transition dynamics in a shallow, microwave-modulated optical lattice, in which center-of-mass wave-packet expansion and tunneling must be considered. Relevance to NASA and relevance of microgravity: High-precision measurement of fundamental constants represents an important part of contemporary atomic-physics research. The proposed high-precision measurement of the Rydberg constant adds a new component to the NASA Fundamental Physics program, and to the Cold Atom Laboratory (CAL) research portfolio. To achieve optimal spectroscopic performance it is critical to reduce systematic lattice-induced transition shifts and related shifts. Since the most critical shifts diminish with decreasing optical-lattice depth, the lattices must be chosen as shallow as possible. Microgravity conditions are important because they enable the use of shallow lattices without the atoms falling out. The microgravity environment provided by the CAL/ its successor instrument, including its 850 nm optical-lattice laser, offers an ideal platform for the project to reach its ultimate uncertainty goal. An International Space Station (ISS)-based setup could also provide a sufficient period of measurement time to acquire good statistics in the high-precision spectra. More generally, the research could bring Rydberg atoms into space. This has far-reaching additional potential for the development of novel quantum technologies for microwave / THz / thermal-field sensing and imaging, and well as for exploration of the physics of very dilute, but strongly interacting Bose Einstein Condensates.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	High-precision spectroscopy and metrology impacts fields such as time measurement and navigation. Benefits to the general public include atomic clocks and the GPS system. The project "High-Precision Microwave Spectroscopy of Long-Lived Circular-State Rydberg Atoms in Microgravity" involves the development of a fundamentally new type of spectroscopy, which has been successfully demonstrated during the first year of the grant. The future development of laser systems to hone this innovative technique may lead to benefits in other disciplines, such as chemistry, biology, and engineering. The research involves graduate students, undergraduate students, and post-docs, who are being prepared for future jobs in academia, government, and industry. Having a capable scientific workforce is essential for progress and competitiveness in science and technology, as well as for the national economy. The main goal of the project, which is to measure the Rydberg constant with circular Rydberg atoms, will help resolve the "proton radius puzzle." The puzzle has arisen from inconsistencies in measurements on low-lying electronic and muonic states of hydrogen, which involve the electromagnetic interaction, QED, the proton structure, and QCD. The project will therefore serve as an indirect test for QED and the structure of the proton. With Rydberg atoms, the Rydberg constant will be tested at electron energies in the range of milli-electronvolts. This regime is rather different from the energy regime of previous measurements (eVs to keVs). More generally, compact two-frequency laser sources can bring Rydberg atoms into space, which carries the potential for novel quantum technologies for microwave / THz / thermal-field sensing and imaging, and well as for exploration of new phases of matter in very dilute, but strongly interacting, Rydberg-dressed Bose Einstein Condensates.
	A measurement of the Rydberg constant using cold circular Rydberg atoms is proposed. Since these atoms have long lifetimes, negligible quantum electrodynamics and no nuclear overlap level shifts, the measurement can shed new light on the "proton radius puzzle." The atoms are trapped in an optical lattice. Transitions are driven using a lattice-modulation technique to perform Doppler-free sub-THz spectroscopy; the transition frequencies yield the Rydberg constant. The selected transitions are free of first-order Zeeman and Stark shifts. Systematic uncertainties due to lattice-induced shifts, core-polarization, second order Zeeman and Stark shifts, etc., must all be estimated and controlled. The measurement will be most accurate and precise in the shallow-lattice, low-temperature environment afforded by microgravity. A quantum model of transitions driven by the lattice-modulation technique must be developed to accurately model spectroscopic lines in this quantum regime. Experiment (1) The cryogenic spectroscopic setup was used to perform measurements of the hyperfine structure splittings of nS1/2 Rydberg states of 85-Rb for principal quantum numbers n=43, 44, 45, and 46. We have obtained Fourier-limited line shapes and determined the line-centers of the hyperfine levels F=2 and F=3 to within 2 kHz or below. The hyperfine splitting is given by A_HFS/(n^*)^-3, with the effective principal quantum number n^* and the hyperfine coupling constant, A_HFS. From the measured splittings, the hyperfine coupling constant A_HFS has been determined to be 15.372(84) GHz. This result is an order-of-magnitude improvement from previous measurements from other groups. We

Task Progress:

have studied and accounted for systematic uncertainty sources, such as unwanted electric and magnetic fields, van der Waals interactions, and AC shifts. A manuscript is in preparation.

(2) To allow for the above measurement, the three-photon excitation scheme was slightly modified from our earlier works. For the hyperfine measurement, we used an excitation scheme 5S F=3 to 5P3/2 F=4 (780 nm), and 5P3/2 F=4 to 5D5/2 F=* (776 nm), and 5D5/2 F=* to nP3/2, F=2 to 4 (1260 nm), where F refers to the respective hyperfine quantum numbers. We have then used microwaves to drive the nP3/2 to nS1/2 (F=2 and F=3) transitions. There, the P Rydberg state has negligible hyperfine structure, while the nS1/2 hyperfine structure is well resolved and allows us to measure A_HFS. To be able to access the nS1/2 F=2 level, the 5P3/2-5D5/2 (776 nm) was driven off-resonantly by about 100 MHz. In this way, the F=2, 3 and 4 states of nP3/2 could be populated. This, in turn, allows microwave transitions into both F-states of nS1/2, enabling measurements of the hyperfine splitting. The method also has the advantage that the 5D5/2 state does not become strongly populated, which leads to a reduction of background signal from Penning ionization of 5D5/2 atoms.

(3) Zeroing of E and B fields in the spectroscopy enclosure. We developed protocols to reduce the background electric and magnetic fields to 0.2 mV/cm and 2 mG using microwave spectroscopy of Rydberg states. Microwave transitions between Rydberg states are much narrower than optical transitions (on the order of 10 kHz vs several MHz), allowing a much improved field zeroing. This level of field zeroing was required for the hyperfine structure measurement, and it also is a prerequisite for circular-state production and spectroscopy.

Theory

(4) We have developed theoretical models for optical circularization of Rydberg atoms using three schemes based on amplitude-modulated optical lattices. In the first method, the Rydberg atoms are circularized in a single (first-order) interaction step by placing them in the field of two Laguerre-Gaussian beams with opposite winding numbers and a frequency difference that equals the transition energy between the initial, low-angular-momentum state (which can be an S, P, or D state) and the desired circular state. The transition frequency is in the order of a THz. The manuscript includes a proposal how this setup could be realized.

In the second method, the atoms are circularized by a sequence of steps akin to the standard rapid adiabatic passage method. Each individual step is a quadrupolar transition effected by an amplitude-modulated ponderomotive optical lattice that increases angular momentum by two units. Since the required modulation frequency is only on the order of a few 100 MHz, the method should be easy to implement.

The third method is an extension of the second in which two orthogonal lattices are phase-modulated in a way that leads to a time-orbiting, sub-micron-sized ponderomotive potential well with an orbiting frequency of several 100 MHz. The resultant poderomotive transitions driven in the atoms are equivalent to transitions driven by a purely circularly polarized RF field. Such fields yield the most robust results in Rydberg-atom circularization, as has been shown in Haroche's group.

(5) Modeling of lattice modulation spectroscopy in the ultracold regime

The experimental work is accompanied by a theoretical effort lead by the co-Principal Investigator (PI) V. Malinovsky, with some associated modeling work provided by the PI. The present objective of the theory work is to model the spectroscopic line shapes of Rydberg transitions in an amplitude-modulated Rydberg-atom lattice in a manner that takes the quantization of the center-of-mass motion into account. This ability is an important ingredient to determine the line center with 1Hz uncertainty, as required in the present project. To take into account the quantization of motion of the Rydberg atoms in the optical lattice, a theoretical model has been developed in which both ground and excited wave functions are subject to the periodic potentials, representing the lattice fields, while the coupling (the effective Rabi frequency) is also periodic as function of the translational coordinate. To obtain the spectrum of the excited-state population, the time-dependent Schroedinger equation has been solved in the momentum representation. The spectra are also modeled using Fermi's Golden Rule on transitions between thermally populated spinor Bloch states in the optical lattice. The results are all in place. Current work is still focused on finalizing two manuscripts on the theory work. The emphases are to (a) make it very clear why this is a new method of Doppler-free spectroscopy, (b) emphasize the utility and the limits when applied in very shallow lattices afforded under microgravity conditions, and (c) to discuss the portability to other spectral regions and molecular-size domains (such as molecules in x-ray lattices that are modulated at optical frequencies).

(6) A paper on three-photon electromagnetically induced transparency (EIT) in a vapor cell has been submitted. The work was conducted by Kaitlin Moore (NASA-funded), an additional National Science Foundation-funded graduate student, and two undergraduate students. This work was loosely related to the NASA-funded work in that some equipment end considerable expertise was shared.

Students

(7) Another graduate student has graduated from the project. After Kaitlin Moore graduated in 2018, this year Andira Ramos has graduated. The training of the new student (Ryan Cardman) has been completed.

Conference presentations

(8) The work was presented at the following national and international conferences (1) the annual NASA fundamental physics conference, (2) the ASGSR (American Society for Gravitational & Space Research) conference in Bethesda, Maryland, the annual Division of Atomic, Molecular and Optical Physics (DAMOP) conference of the American Physical Society (APS), and the CPEM 2018 in Paris, France (Conference on precision electromagnetic measurements). This conference was focused on fundamental constants and the redefinition of the SI (International System of Units), and had a session on the proton radius.

Bibliography Type: Description: (Last Updated: 02/24/2025)

Papers from Meeting Proceedings

Raithel G, Ramos A, Moore K, Malinovsky V. "Measurement of the Rydberg Constant with Trapped Rydberg Atoms." 2018 CPEM Conference on Precision Electromagnetic Measurements, Paris, France, July 8-13, 2018.
In: 2018 Conference on Precision Electromagnetic Measurements (CPEM 2018).
https://doi.org/10.1109/CPEM.2018.8501136 , Sep-2018